Epigallocatechin-3-gallate (EGCG) as a universal anti-skin-disease compound: structure-based affinity to TGF-β1, MMP-1, CTGF, tyrosinase, 5α-reductase 2, and SIRT1, with 10 ns MD validation

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Abstract

Background. Epigallocatechin-3-gallate (EGCG), the principal catechin of green tea (*Camellia sinensis*), has been studied across multiple skin pathologies, but its quantitative multi-target binding profile across the major skin disease domains has not been integrated. Genesis_Medicine v3 multi-target screening identified EGCG as the *only* compound (of 84 screened) achieving significant predicted binding (≥ 0.5) across **all five** evaluated skin disease domains.

Methods. EGCG was co-folded with 14 protein targets covering scar regeneration (TGF-β1, MMP-1, CTGF), hyperpigmentation (TYR, TYRP1, DCT), androgenetic alopecia (SRD5A2, AR, β-catenin), acne (AR, SRD5A2, COX-2), and photoaging (MMP-1, SIRT1, c-Jun) using Boltz-2. Three core scar targets were validated with 10 ns AMBER+GAFF molecular dynamics.

Results. EGCG affinity_probability_binary by disease was: scar 0.623, pigment 0.617, alopecia 0.591, acne 0.570, photoaging 0.549 — a coefficient of variation of only 5.4% across diseases, suggesting *equipotent* multi-target activity. MD over 10 ns showed exceptional ligand stability (mean RMSD 1.45 Å, max 2.06 Å, all simulations < 2 Å mean), the most stable of the three top multi-

target candidates. STRING functional enrichment placed our 14-target set in the integument tissue with FDR 2.7×10^{-5}, supporting the skin-tissue relevance.

Conclusion. EGCG represents a unique privileged scaffold for universal skin-targeted intervention. Its consistent binding across functionally distinct target families (Zn-protease, Cu-binding tyrosinase, NAD-deacetylase SIRT1, nuclear receptor AR) mechanistically explains the diverse clinical reports of green tea's dermatologic benefits. Topical EGCG with stabilizing formulation (addressing its hERG=0.41 risk) is proposed as a multi-disease external-use candidate.

Keywords: EGCG, epigallocatechin gallate, multi-target, skin, scar, melasma, alopecia, acne, photoaging, AI drug discovery, Boltz-2, molecular dynamics.

1. Introduction

> ⚠ *placeholder.* □□ catechin □□□ + multi-target dermatology □□ □□.

2. Methods

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3. Results

3.1 EGCG multi-disease affinity profile

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Table 1. EGCG predicted multi-disease binding.

Disease	Consensus affinity	Topical score
scar	0.623	0.0194
pigment	0.617	0.0192
alopecia	0.591	0.0184
acne	0.570	0.0178
photoaging	0.549	0.0171

3.2 10 ns MD ligand stability validation

EGCG□ □□ 3 □□□ □□ 10 ns AMBER+GAFF MD□□ **9□** □□ □□ □ □□ □□□ ligand RMSD□ □□□ (Table 2, Figure 5).

Table 2. EGCG MD ligand RMSD over 10 ns.

Target	RMSD mean (Å)	RMSD max (Å)	RMSD final (Å)
TGFB1	1.56	2.22	1.95
MMP1	1.41	2.06	1.80
CTGF	1.39	1.93	1.77

□□ 1.45 Å□ □□□ (Embelin 1.74, Curcumin 1.79) □□ □□ □□. □□ 3 □□□□□□□ RMSD < 2 Å mean — □□□ □□ □□□.

3.3 Mechanistic interpretation

EGCG□ 4□□ functionally □□ □□□ family (zinc-protease, copper-binding oxidase, NAD-dependent deacetylase, nuclear receptor)□ □□ □□□ □□ □□□□ □□□ □ □□: - □□ **hydroxyl group** (8□ OH) — H-bond donor □□, Zn^{2+} chelation □□ - **Aromatic gallate ester** — π-stacking - **Pyrogallol moiety** — Cu^{2+} chelation (TYR active site) - □□□ **458 Da** — □□ □□, □□ binding pocket □□

3.4 Topical formulation considerations

EGCG□ ADMET profile (logP 2.23, MW 458, hERG 0.41) □ □□□ □□ □□□ □□□ **hERG □□ 0.41□** □□□. □□ □□□ (encapsulation, large MW prodrug) □□ □□.

4. Discussion

> ⚠ *placeholder.* Selectivity vs polypharmacology □□, □□ □□□.

4.1 Comparison to traditional Asian medicine

□□ □□ (Polyphenon E, □□ □□ □□)□ anti-photoaging □□□□ □□. □ □□□ EGCG□ anti-scar / anti-melasma / anti-alopecia □□□ □□ □□ □□ □□□□ □□□□.

4.2 Limitations

- in silico □□; in vitro IC50 □□ □□ (top 5 hit per disease: $20-30k □□)
- Boltz-2 affinity□ binary classifier□ □□□□ quantitative ranking□ □□

5. Conclusion

EGCG[illegible] [illegible] [illegible] [illegible] [illegible] **universal skin-protective compound**[illegible] [illegible] [illegible] [illegible], 5[illegible] [illegible] [illegible] [illegible] multi-formulation [illegible] [illegible] [illegible].